

Seifeddine Reguige

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SUMMARY

Computer Science Co-op student graduating in August 2026 with hands-on experience in data science, software development, research infrastructure, databases, and web/mobile applications. Strong foundation in Python, Java, SQL, machine learning, data pipelines, REST APIs, and full-stack development, with applied projects in recommender systems, computer vision, Android systems, and secure data platforms.

EDUCATION

University of Ottawa

Ottawa, ON

Bachelor of Science in Computer Science (Co-op)

Expected Graduation: Aug. 2026

- Dean's Honour List / very good academic standing; relevant coursework in Artificial Intelligence, Data Science, Databases, Operating Systems, Software Engineering, Data Structures and Algorithms, Computer Networks, and Secure Systems Design.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, PHP, SQL, Bash, HTML/CSS, XML

Data Science & ML: Pandas, NumPy, scikit-learn, PyTorch, TensorFlow, regression, clustering, recommender systems, feature engineering, model evaluation, data visualization

Backend & Databases: REST APIs, Supabase, SQLite, relational data modeling, ETL pipelines, CSV/JSON/XML processing, metadata documentation, data quality validation

Software & Tools: Git, GitHub, Docker, Streamlit, Jupyter, VS Code, Android Studio, Chrome DevTools, Linux/Unix, technical documentation

Computer Vision: OpenCV, optical flow, RAFT, Farneback, Endpoint Error (EPE), motion metadata analysis

IT Support: workstation troubleshooting, system configuration, assistive technology, monitoring software, hardware/software diagnostics

EXPERIENCE

Research Assistant / Data Scientist

Ottawa, ON

University of Ottawa – Faculty of Social Sciences

Summer 2026 – Present

- Contributing to the development of a secure digital research infrastructure for sensitive public health and policy research data, supporting collection, organization, analysis, and controlled data management workflows.
- Supporting legacy data migration and dataset structuring through CSV exports, data cleaning, normalization, metadata documentation, quality validation, and preparation of analysis-ready datasets.
- Collaborating with researchers, technical staff, and infrastructure teams to define data models, document variables, organize scholarly resources, and support future relational database integration.
- Applying computer science and data science practices to improve data governance, secure storage, research documentation, and future analysis and visualization workflows.

IT Systems Developer

Ottawa, ON

Faculty of Health Sciences, University of Ottawa

Oct. 2024 – Apr. 2025

- Developed and maintained web-based systems and data-driven interfaces using Java, PHP, JavaScript, HTML, and CSS for faculty platforms serving 10,000+ monthly users.
- Worked with backend logic, API interactions, and structured data processing to support research and operational workflows across academic and healthcare environments.
- Built responsive interfaces, optimized front-end performance through JavaScript refactoring and debugging, and improved usability for faculty members, staff, and researchers.
- Collaborated with stakeholders to gather requirements, troubleshoot technical issues, document workflows, and deliver maintainable system improvements.

PROJECTS

Metadata-Based Optical Flow Reliability Predictor

Python, PyTorch, TensorFlow, OpenCV, scikit-learn, Streamlit

- Built an end-to-end machine learning pipeline to predict optical-flow reliability from scenario metadata using TFRecord-based video datasets and engineered motion features from camera/object dynamics.
- Evaluated Farneback and RAFT error labels using Endpoint Error, then compared Linear Regression, Random Forest, and Gradient Boosting under scenario-holdout validation.
- Created a Streamlit explorer with plots, exported media, flow visualizations, EPE heatmaps, and qualitative analysis tools for model interpretation.

Movie Recommender System

Python, Pandas, NumPy, scikit-learn, Surprise

- Built and compared content-based filtering, clustering, and collaborative filtering approaches on The Movies Dataset and user-item rating data.
- Engineered sparse utility matrices for memory-efficient matrix factorization and evaluated recommendation relevance using SVD, MSE, Precision@K, and Mean Reciprocal Rank.
- Applied Cosine and Jaccard similarity with KMeans and DBSCAN to capture user/item preference patterns and reduce popularity bias.

PCSurMeasure: Role-Based Mobile Ordering System

Java, Android Studio, Supabase, REST APIs

- Led development of a multi-role Android ordering application with authentication, permission-controlled workflows, order lifecycle management, and API-driven backend interactions.
- Migrated local SQLite storage to Supabase and applied MVC architecture to improve scalability, maintainability, and separation of concerns.

Traveling Salesman Problem Heuristics Visualizer

C++, Python, Flask, Algorithms, Data Visualization

- Implemented and compared constructive and local-search heuristics, including Nearest Neighbor, Insertion, Random, 2-OPT, and 3-OPT, for approximate TSP optimization.
- Built a lightweight visualization workflow to export city coordinates and route results, enabling comparison of path quality and algorithm behavior.